Lab 47: Nightly job on Batches, upload flow stays synchronous

Topic 4: Prompt Engineering & Structured Output · Task 4.5: Design efficient batch processing strategies
Runs on Simulator 6: Structured Data Extraction · Self-practice
Source: SkillCertPro practice exam Part 1, Q32 (unofficial third-party question)

The question

This team has two extraction workloads: an interactive upload flow where users wait for validation before submitting documents, and a nightly archive-reprocessing job whose results are reviewed the next business day. Leadership wants to reduce token costs without degrading the user-facing flow. What should the architect recommend?

A. Route the nightly archive reprocessing through the Message Batches API and keep interactive validation on synchronous Messages API calls.

B. Route interactive validation through the Message Batches API and reserve synchronous calls for nightly archive reprocessing.

C. Keep both workloads on synchronous Messages API calls and avoid batch because results may take the full processing window.

D. Route both workloads through the Message Batches API and rely on typical one-hour completion for interactive validation.

Simulate the solution

```
# In the browser: open Simulator 6, then Lab 47
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 47

# In a terminal, from simulators/simulator-6-structured-extraction/
uv run extract.py --case UC-32
```

What you should see

interactive_upload routes to sync, nightly_archive to batch, with a 0.5 cost multiplier.

evidence/cases/UC-32.json routes interactive_upload to sync and nightly_archive to batch, records the sync latency in seconds, the batch id, and the 0.5 cost multiplier for the batch workload.

Open `simulators/simulator-6-structured-extraction/evidence/` and find it yourself.

The answer

A. Route by who is waiting.

Batches trade latency for cost: 50% of standard prices, but a batch can take up to 24 hours. Send latency-tolerant work (nightly reprocessing, offline audits, next-day reports) to batch, and keep blocking user flows synchronous. “Most batches complete within 1 hour” is not a service guarantee.

Why the others fail

B. Exactly backwards: it puts users behind a batch window of up to 24 hours.

C. Gives up the 50% saving on a workload that doesn't need fast results.

D. Typical completion time is not a guarantee, so users could wait hours.

Check yourself

- Learner justifies each route by who is waiting and by when the result is needed.
- Learner states the two batch trade-offs (price, processing window) without quoting the 1-hour figure as a guarantee.
- Learner explains why the SLA for a nightly job still needs a submission cut-off time.

Now change one thing

Change DOC-04's printed total in fixtures/invoices.txt, rerun, and watch by_document_type.invoice fall while the others hold.

Third-party practice material from SkillCertPro, reconstructed from a video transcript in the Tertiary Markdown Vault. Not a real exam item. Wording may differ from the original.